Stochastic Whale Model Memo

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Abstract

This document describes a stochastic whale model developed for simulations of autonomous underwater vehicles (AUVs) detecting whale calls. Whales move and call while crossing a horizontal monitoring line represented in local north–east coordinates. The model represents individual variation, group-level dependence, temporally correlated movement, and temporally correlated calling activity. The current acoustic-detection component is a binary range model and is explicitly treated as a baseline to be replaced by a received-level and detector-performance model. Parameter values listed here are current simulation settings; entries marked "source required" are assumptions that still require species-, population-, season-, and behavior-specific support.

1 Monitoring-line geometry

Let the start and end of the monitoring line be

$$\mathbf{x}_s = (n_s, e_s), \qquad \mathbf{x}_e = (n_e, e_e), \tag{1}$$

where n and e denote local north and east coordinates. The line displacement and length are

$$\Delta\mathbf{x} = \mathbf{x}_e - \mathbf{x}_s, \qquad L = \|\Delta\mathbf{x}\|_2. \tag{2}$$

The along-line and perpendicular unit vectors are

$$\hat{\mathbf{a}} = \frac{\Delta\mathbf{x}}{L}, \qquad \hat{\mathbf{p}} = (-\hat{a}_e, \hat{a}_n). \tag{3}$$

A zero-length line is invalid because it cannot define either direction.

2 Group formation and individual assignment

For N whales and configured mean group size $\bar{G}$, the number of groups is

$$N_G = \max\left(1, \left\lceil \frac{N}{\bar{G}} \right\rceil\right). \tag{4}$$

Whale $i \in \{0, \ldots, N-1\}$ is assigned in round-robin order to

$$g(i) = i \bmod N_G. \tag{5}$$

This numerical rule keeps realized group sizes approximately equal; it is not an empirical group-size distribution.

3 Crossing location and activation time

Each group g is assigned a nominal crossing distance drawn uniformly along the monitoring line,

$$D_g \sim \mathcal{U}(0, L), \tag{6}$$

and a nominal activation time drawn uniformly over the mission,

$$T_g \sim \mathcal{U}(0, T_{\text{mission}}). \tag{7}$$

The group's starting side is selected with equal probability:

$$\Pr(s_g = +1) = \Pr(s_g = -1) = \frac{1}{2}. \tag{8}$$

An individual crossing distance is drawn from a normal distribution centered on the group value:

$$D_i \sim \mathcal{N}(D_{g(i)}, \sigma_D^2), \qquad 0 \le D_i \le L. \tag{9}$$

Samples outside $[0, L]$ are rejected, producing a truncated normal distribution rather than clipped values. The corresponding crossing point is

$$\mathbf{x}_i^{\text{cross}} = \mathbf{x}_s + D_i \hat{\mathbf{a}}. \tag{10}$$

The individual activation time is drawn from a normal distribution centered on the group value:

$$T_i \sim \mathcal{N}(T_{g(i)}, \sigma_T^2), \qquad 0 \le T_i < T_{\text{mission}}, \tag{11}$$

with invalid samples rejected.

4 Initial position and swimming speed

Let $s_g \in \{-1, +1\}$ identify the group's starting side of the monitoring line, and let d_0 be the configured starting distance. The initial position is

$$\mathbf{x}_{i,0} = \mathbf{x}_i^{\text{cross}} + s_g d_0 \hat{\mathbf{p}}. \tag{12}$$

Individual swimming speed is drawn from a positive-truncated normal distribution,

$$V_i \sim \mathcal{N}(\mu_V, \sigma_V^2), \qquad V_i > 0, \tag{13}$$

implemented by rejecting nonpositive draws. The nominal crossing velocity points toward the line:

$$\mathbf{v}_i^{\text{cross}} = -s_g V_i \hat{\mathbf{p}}. \tag{14}$$

With headings measured clockwise from north, the fixed crossing heading is

$$\theta_i^{\text{cross}} = \operatorname{atan2}\left(v_{i,e}^{\text{cross}}, v_{i,n}^{\text{cross}}\right). \tag{15}$$

5 Correlated movement

Each whale has a fixed crossing heading θ_i^{cross} that initially points from its starting position toward the monitoring line. The whale's actual heading varies around this crossing direction through correlated group-level and individual-level heading errors.

Heading errors are updated every Δt_H seconds. Let $G_{g,k}$ denote the heading error shared by group g during interval k. It follows the AR(1) process

$$G_{g,k} = \rho_H G_{g,k-1} + \sigma_{H,\epsilon}\epsilon_{g,k}, \qquad \epsilon_{g,k} \sim \mathcal{N}(0,1), \tag{16}$$

where ρ_H controls temporal persistence. The innovation standard deviation is

$$\sigma_{H,\epsilon} = \sigma_H\sqrt{1-\rho_H^2}, \qquad \sigma_H = \sigma_{H,\text{deg}}\frac{\pi}{180}. \tag{17}$$

This scaling gives the stationary group heading-error process a standard deviation of σ_H.

Each whale also has an independent heading-error process,

$$I_{i,k} = \rho_H I_{i,k-1} + \sigma_{H,\epsilon}\eta_{i,k}, \qquad \eta_{i,k} \sim \mathcal{N}(0,1). \tag{18}$$

Let $f_H \in [0,1]$ denote the configured shared-heading fraction. The total heading error for whale i in group g is

$$E_{i,k} = \sqrt{f_H}\, G_{g,k} + \sqrt{1-f_H}\, I_{i,k}. \tag{19}$$

Because the group and individual processes are independent, the square-root weights preserve the configured heading-error variance once both processes have reached stationarity. Initialization at zero creates an initial transient period with smaller variance.

The whale's actual heading during interval k is

$$\theta_{i,k} = \theta_i^{\text{cross}} + E_{i,k}. \tag{20}$$

Thus, the heading fluctuates around the original crossing direction rather than accumulating successive turns into an unrestricted heading random walk. The group process begins at zero at mission time $t = 0$, whereas each whale's individual process begins at zero when its trajectory is generated. Because a whale may activate after the first group heading interval, its initial realized heading can already contain a nonzero group-level deviation.

Group heading updates are aligned to absolute mission-time boundaries. Consequently, a whale that activates between two boundaries has a shorter first movement interval, after which its trajectory uses the regular interval duration Δt_H.

The north and east velocity components during the interval are

$$v_{i,n,k} = V_i \cos\theta_{i,k}, \tag{21}$$

$$v_{i,e,k} = V_i \sin\theta_{i,k}. \tag{22}$$

Velocity is constant between heading updates. If the duration from the current time to the next heading-update boundary is Δt_k, the position is propagated according to

$$n_i(t+\Delta t_k) = n_i(t) + v_{i,n,k}\Delta t_k, \tag{23}$$

$$e_i(t+\Delta t_k) = e_i(t) + v_{i,e,k}\Delta t_k. \tag{24}$$

Equivalently,

$$n_i(t+\Delta t_k) = n_i(t) + V_i\cos\theta_{i,k}\Delta t_k, \tag{25}$$

$$e_i(t+\Delta t_k) = e_i(t) + V_i\sin\theta_{i,k}\Delta t_k. \tag{26}$$

6 Calling model

6.1 Individual baseline rate

The current implementation draws an individual baseline rate from configured limits,

$$\lambda_i^{(0)} \sim \mathcal{U}(\lambda_{\min}, \lambda_{\max}), \tag{27}$$

in calls per whale per hour. This uniform distribution is provisional. A lognormal, gamma, empirical, or behavioral-state distribution may be more appropriate once a target species and context are selected.

6.2 Group-level calling activity

Whales in the same group share a time-varying calling-activity multiplier. Calling activity is divided into intervals of duration Δt_A. Let $A_{g,k}$ denote the latent calling-activity state of group g during interval k.

The initial state is drawn from the stationary distribution,

$$A_{g,0} \sim \mathcal{N}(0,1). \tag{28}$$

Subsequent states follow the AR(1) process

$$A_{g,k} = \rho_A A_{g,k-1} + \sqrt{1-\rho_A^2}\,\epsilon_{g,k}, \qquad \epsilon_{g,k} \sim \mathcal{N}(0,1), \tag{29}$$

where ρ_A is the configured call-activity correlation. The factor $\sqrt{1-\rho_A^2}$ ensures that the stationary variance of $A_{g,k}$ remains equal to one.

The latent activity state is transformed into a positive call-rate multiplier using

$$M_{g,k} = \exp\left(\sigma_G A_{g,k} - \frac{\sigma_G^2}{2}\right), \tag{30}$$

where σ_G is `group_log_rate_std`. Because $A_{g,k} \sim \mathcal{N}(0,1)$ at stationarity, subtracting $\sigma_G^2/2$ ensures

$$\mathbb{E}[M_{g,k}] = 1. \tag{31}$$

All whales in group g use the same multiplier during interval k. Therefore, group members tend to experience active and quiet calling periods together. If whale i has baseline call rate $\lambda_i^{(0)}$, its effective rate during interval k is

$$\lambda_{i,k} = \lambda_i^{(0)} M_{g(i),k}. \tag{32}$$

6.3 Call-event generation

Conditional on $\lambda_{i,k}$, calls are generated as a piecewise-homogeneous Poisson process. For an interval of duration τ hours,

$$N_{i,k} \mid \lambda_{i,k} \sim \text{Poisson}(\lambda_{i,k}\tau). \tag{33}$$

After drawing $N_{i,k}$, the model independently places each call uniformly within the activity interval:

$$C_{i,k,m} \mid N_{i,k} \sim \mathcal{U}(t_k, t_{k+1}), \qquad m = 1, \ldots, N_{i,k}. \tag{34}$$

The generated call times are then sorted chronologically. Drawing a Poisson count and conditional uniform event times is equivalent to simulating a homogeneous Poisson process within each interval. Calls are generated only between the whale's activation time and the final time of its corridor trajectory.

7 Lifecycle

Before T_i, the whale remains at its initial position with zero published velocity. For $T_i \le t < T_i^{\text{end}}$, its state is interpolated from the generated trajectory. At or after T_i^{end}, the whale is marked complete and is no longer treated as traversing the monitored corridor.

8 Current detection baseline

The present model declares a call detected by vehicle j when the horizontal separation is within a fixed radius R_D:

$$y_{ij}(t) = \begin{cases} 1, & \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|_2 \le R_D, \\ 0, & \text{otherwise.} \end{cases} \tag{35}$$

This is an engineering baseline, not a realistic acoustic model. A physical model should calculate received level or signal-to-noise ratio from call source level, propagation loss, ambient noise, vehicle self-noise, receiver response, and detector performance. The detector should ultimately be characterized by an empirical probability of detection and false-alarm rate.

9 Current parameterization and evidence status

Parameter	Code/model representation	Current value	Source or status
Number of whales	N; `whale_count`	46 whales	?
Mission duration	T_{mission}	3 days	?
Monitoring-line length	L	126 km	Distance between Cox's Ledge and Nantucket Shoals
Whale depth	`whale_depth`	20 m	?
Mean group size	$\bar{G}$; `group_mean_size`	3 whales	?
Group assignment	$g(i) = i \bmod N_G$	Round-robin	Numerical balancing rule, not an empirical distribution.
Group-position spread	σ_D; `group_position_std`	1000 m	?
Group start-time spread	σ_T; `group_start_time_std`	600 s	?
Starting distance	d_0; `start_distance`	2000 m	?
Mean speed	μ_V; `speed_mean`	1.5 m s^{-1}	Avg. feeding speed
Speed standard deviation	σ_V; `speed_std`	0.2 m s^{-1}	?
Heading-update interval	Δt_H; `turn_interval`	60 s	?
Heading correlation	ρ_H; `heading_correlation`	0.95	?
Heading standard deviation	$\sigma_{H,\text{deg}}$	15°	?

Parameter	Code/model representation	Current value	Source or status
Shared-heading fraction	f_H; `shared_heading_fraction`	0.75	?
Baseline call rate	$\lambda_i^{(0)}$	$\mathcal{U}(4.8, 8.0)$ calls whale^{-1} h^{-1}	?
Mean baseline call rate	$\mathbb{E}[\lambda_i^{(0)}]$	6.4 calls whale^{-1} h^{-1}	?
Group log-rate spread	σ_G; `group_log_rate_std`	0.50	?
Activity interval	Δt_A; `call_activity_interval`	600 s	?
Activity correlation	ρ_A; `call_activity_correlation`	0.90	?
Call-event process	Conditional Poisson process	Piecewise rate	?
Detection radius	R_D	11 km	Source
Detector probability	Binary inside/outside R_D	1/0	?

10 Known limitations

1. The model is not yet parameterized for a single defined species, population, season, and behavioral state.
2. Round-robin assignment enforces balanced groups rather than sampling an observed group-size distribution.
3. Several movement and calling parameters are currently assumptions rather than fitted quantities.
4. A conditional Poisson process may not reproduce observed overdispersion or long silent periods.
5. Binary range-based detection omits source-level variation, propagation, noise, receiver response, false alarms, and missed detections.
6. A detected call does not establish the operational identity of a unique whale. "Whales detected" is presently a simulation ground-truth coverage metric.

References

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